

BUSSELTON MARGARET RIVER, AustraliaWMO: **956110**Lat: **33.6858S**Lon: **115.4008E**Elev: **17**StdP: **101.12**Time Zone: **8.00 (E08)**Period: **97-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.4	4.4	1.2	4.1	9.0	2.4	4.5	8.6	12.2	15.3	11.2	15.3	1.9	100	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.9	34.0	19.4	32.2	19.1	30.5	18.6	21.2	29.2	20.4	28.2	19.7	27.2	5.1	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.0	22.5	18.1	13.0	21.9	17.2	12.3	21.6	61.6	29.5	58.6	28.1	56.2	27.2	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	9.5	8.7	1.3	37.5	0.7	1.1	0.8	38.3	0.4	38.9	0.0	39.5	-0.5	40.3		
			1.0	23.0	0.7	1.0	0.5	23.6	0.1	24.2	-0.3	24.7	-0.9	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.3	21.7	21.8	20.2	17.2	14.4	12.5	11.6	12.0	12.6	14.5	17.6	19.8
	DBStd	4.59	3.08	3.18	3.19	2.79	2.20	2.04	2.02	1.98	2.15	2.58	3.11	3.45
	HDD10.0	18	0	0	0	0	1	2	7	4	3	1	0	0
	HDD18.3	1154	6	6	17	54	122	174	208	197	173	122	51	23
	CDD10.0	2321	364	332	317	216	138	78	57	66	81	140	227	305
	CDD18.3	415	112	104	76	21	2	0	0	0	0	4	28	69
	CDH23.3	4223	1161	1065	713	147	8	0	0	0	3	48	309	769
	CDH26.7	1630	493	447	245	23	0	0	0	0	0	7	97	318
Wind	WSAvg	4.4	5.4	5.3	4.8	3.9	3.5	3.8	3.8	3.8	4.0	4.2	4.7	5.1
Precipitation	PrecAvg	791	8	10	20	42	112	166	164	116	76	44	26	12
	PrecMax	1148	70	64	100	163	245	309	330	231	148	123	81	71
	PrecMin	532	0	0	0	0	34	64	64	35	31	1	1	0
	PrecStd	133	14	16	25	34	54	62	59	44	30	26	19	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.4	35.6	34.5	29.2	24.4	20.3	19.0	19.8	22.8	27.7	32.3	35.5
		MCWB	19.9	20.5	19.8	17.6	15.8	14.9	15.5	15.5	14.9	17.2	17.7	19.4
	2%	DB	33.7	33.4	31.3	26.9	22.1	19.0	17.9	18.3	20.2	24.5	29.6	33.0
		MCWB	19.4	19.7	18.7	17.1	15.7	15.2	14.6	14.5	15.0	16.2	17.3	19.0
	5%	DB	31.7	31.4	29.3	24.9	20.8	18.1	17.0	17.4	18.7	22.2	26.9	30.6
		MCWB	19.1	19.3	18.2	16.9	15.8	14.6	13.9	13.9	14.3	15.5	16.8	18.2
	10%	DB	29.4	29.6	27.3	23.2	19.7	17.3	16.2	16.5	17.5	20.3	24.5	27.7
		MCWB	18.7	18.7	17.9	16.6	15.5	14.0	13.3	13.3	13.7	14.8	16.3	17.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.5	23.0	21.6	20.2	19.2	17.5	17.0	16.9	17.4	18.6	19.7	21.1
		MCDB	30.1	29.5	28.9	23.2	20.5	18.6	17.9	18.2	19.4	24.0	26.8	30.9
	2%	WB	21.1	21.1	20.5	19.2	18.0	16.3	15.7	15.7	16.1	17.4	18.7	20.0
		MCDB	29.5	29.5	27.1	22.7	20.1	17.9	17.0	17.4	18.9	22.3	25.5	29.7
	5%	WB	20.1	20.2	19.7	18.3	17.0	15.4	14.7	14.7	14.9	16.3	17.8	19.0
		MCDB	28.9	28.8	26.0	22.6	19.5	17.3	16.3	16.7	18.0	20.8	24.8	28.2
	10%	WB	19.1	19.4	18.8	17.4	16.0	14.5	13.7	13.7	14.1	15.4	16.9	18.0
		MCDB	27.7	27.7	24.9	22.0	18.9	16.6	15.7	15.9	16.9	19.5	23.6	26.3
Mean Daily Temperature Range	5% DB	MDBR	14.2	13.9	12.9	11.5	10.4	9.2	8.9	9.0	9.6	11.3	12.5	13.7
		MCDBR	17.3	16.4	15.2	13.5	11.8	9.9	9.2	9.7	11.7	14.4	16.1	17.4
		MCWBR	6.4	5.9	5.7	6.1	7.0	6.8	6.6	6.7	7.3	7.4	6.8	6.7
		MCDDBR	14.7	14.0	12.0	10.5	9.5	8.3	8.0	8.4	10.2	11.7	13.7	15.5
	5% WB	MCWBR	6.2	5.8	5.4	6.1	7.0	6.8	6.8	6.7	7.2	7.0	6.7	6.6
Clear-Sky Solar Irradiance	taub	0.333	0.333	0.322	0.316	0.311	0.309	0.303	0.314	0.329	0.327	0.333	0.327	
	taud	2.591	2.600	2.631	2.641	2.624	2.610	2.627	2.585	2.519	2.543	2.548	2.588	
	Ebn at Noon	1004	984	960	912	863	837	863	900	936	979	998	1014	
	Edh at Noon	104	100	90	80	72	68	70	83	99	104	108	105	
All-Sky Solar Radiation	RadAvg	7.84	6.98	5.59	3.98	2.88	2.31	2.48	3.31	4.44	5.89	7.11	7.83	
	RadStd	0.40	0.29	0.29	0.24	0.20	0.15	0.13	0.20	0.25	0.31	0.41	0.39	